

Seminar Nr. 7, Probability and Statistics

Inequalities; Central Limit Theorem; Point Estimators

Theory Review

Markov's Inequality: $P(|X| \geq a) \leq \frac{1}{a} E(|X|), \forall a > 0.$

Chebyshev's Inequality: $P(|X - E(X)| \geq \varepsilon) \leq \frac{V(X)}{\varepsilon^2}, \forall \varepsilon > 0.$

Central Limit Theorem (CLT) Let $X_1, \dots, X_n$ be independent random variables with the same expectation $\mu = E(X_i)$ and same standard deviation $\sigma = \sigma(X_i) = \text{Std}(X_i)$ and let $S_n = \sum_{i=1}^n X_i$. Then, as $n \rightarrow \infty$,

$$Z_n = \frac{S_n - E(S_n)}{\sigma(S_n)} = \frac{S_n - n\mu}{\sigma\sqrt{n}} \rightarrow Z \in N(0, 1), \text{ in distribution (in cdf), i.e. } F_{Z_n} \rightarrow F_Z = \Phi.$$

Point Estimators

- method of moments: solve the system $\nu_k = \bar{\nu}_k$, for as many parameters as needed ($k = 1, \dots$, nr. of unknown parameters);

- method of maximum likelihood: solve $\frac{\partial \ln L(X_1, \dots, X_n; \theta)}{\partial \theta_j} = 0$, where $L(X_1, \dots, X_n; \theta) = \prod_{i=1}^n f(X_i; \theta)$ is the likelihood function;

- **standard error** of an estimator $\bar{\theta}$: $\sigma_{\hat{\theta}} = \sigma(\bar{\theta}) = \sqrt{V(\bar{\theta})}$;

- **Fisher information** $I_n(\theta) = -E \left[\frac{\partial^2 \ln L(X_1, \dots, X_n; \theta)}{\partial \theta^2} \right]$; if the range of X does not depend on θ , then $I_n(\theta) = nI_1(\theta)$;

- **efficiency** of an absolutely correct estimator $\bar{\theta}$: $e(\bar{\theta}) = \frac{1}{I_n(\theta)V(\bar{\theta})}$.

- an estimator $\bar{\theta}$ for the target parameter θ is

- **unbiased**, if $E(\bar{\theta}) = \theta$;
- **absolutely correct**, if $E(\bar{\theta}) = \theta$ and $V(\bar{\theta}) \rightarrow 0$, as $n \rightarrow \infty$;
- **MVUE** (minimum variance unbiased estimator), if $E(\bar{\theta}) = \theta$ and $V(\bar{\theta}) \leq V(\hat{\theta})$, $\forall \hat{\theta}$ unbiased estimator;
- **efficient**, if $e(\bar{\theta}) = 1$.

- $\bar{\theta}$ efficient $\Rightarrow \bar{\theta}$ MVUE.

1. (The 3σ Rule). For any random variable X , most of the values of X lie within 3 standard deviations away from the mean.

2. True or False: There is at least a 90% chance of the following happening: when flipping a coin 1000 times, the number of "heads" that appear is between 450 and 550.

3. Installation of some software package requires downloading 82 files. On the average, it takes 15 sec to download a file, with a variance of 16 sec². What is the probability that the software is installed in less than 20 minutes?

4. A sample of 3 observations, $X_1 = 0.4, X_2 = 0.7, X_3 = 0.9$, is collected from a continuous distribution with pdf

$$f(x; \theta) = \begin{cases} \theta x^{\theta-1}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases},$$

with $\theta > 0$, unknown. Estimate θ by the method of moments and by the method of maximum likelihood.

5. A sample $X_1, \dots, X_n$ is drawn from a distribution with pdf

$$f(x; \theta) = \frac{1}{2\theta} e^{-\frac{x}{2\theta}}, \quad x > 0$$

($\theta > 0$), which has mean $\mu = E(X) = 2\theta$ and variance $\sigma^2 = V(X) = 4\theta^2$. Find

a) the method of moments estimator, $\bar{\theta}$, for θ ;

b) the efficiency of $\bar{\theta}$, $e(\bar{\theta})$;

c) an approximation for the standard error of the estimate in **a)**, $\sigma_{\bar{\theta}}$, if the sum of 100 observations is 200.